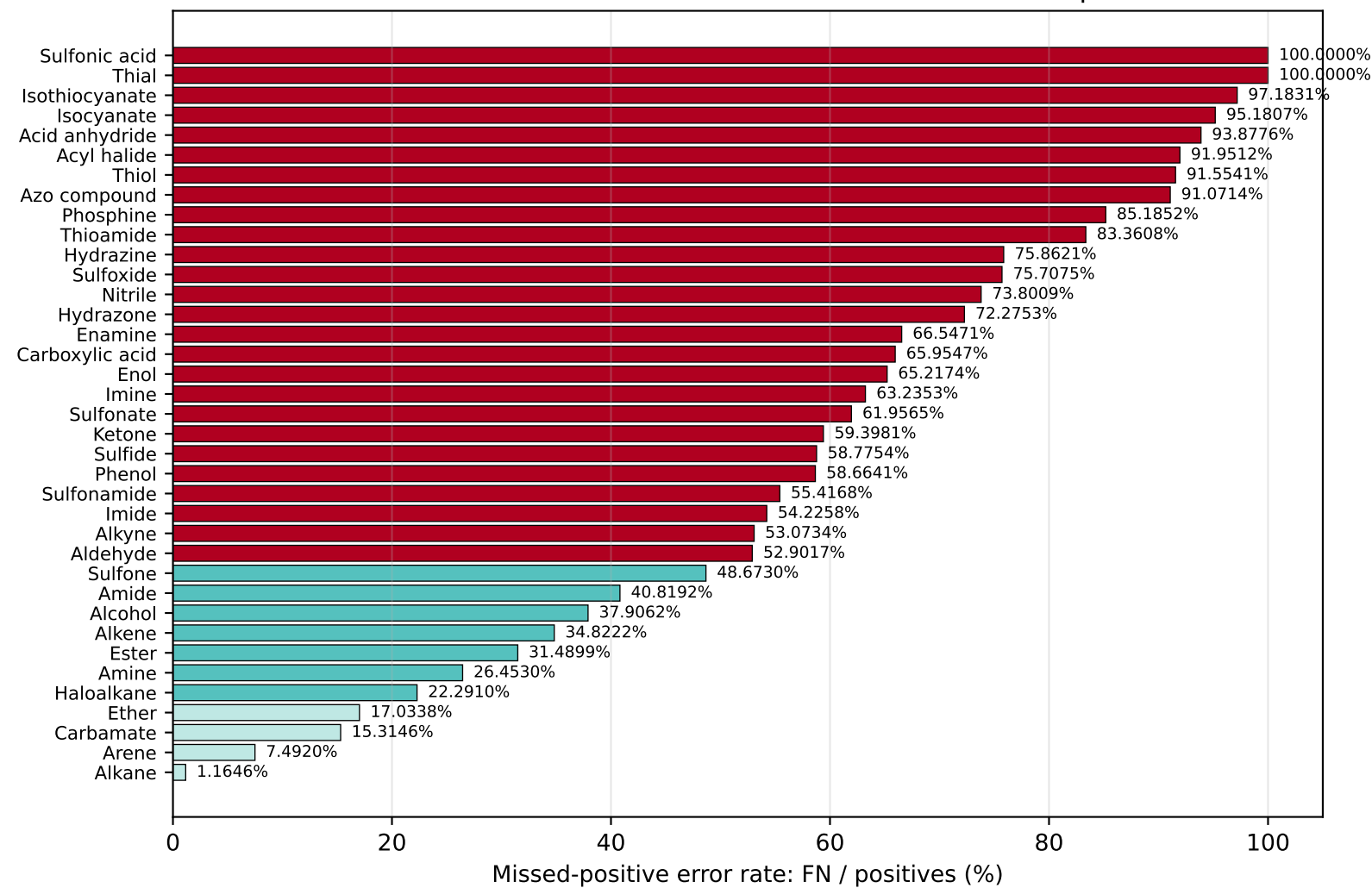
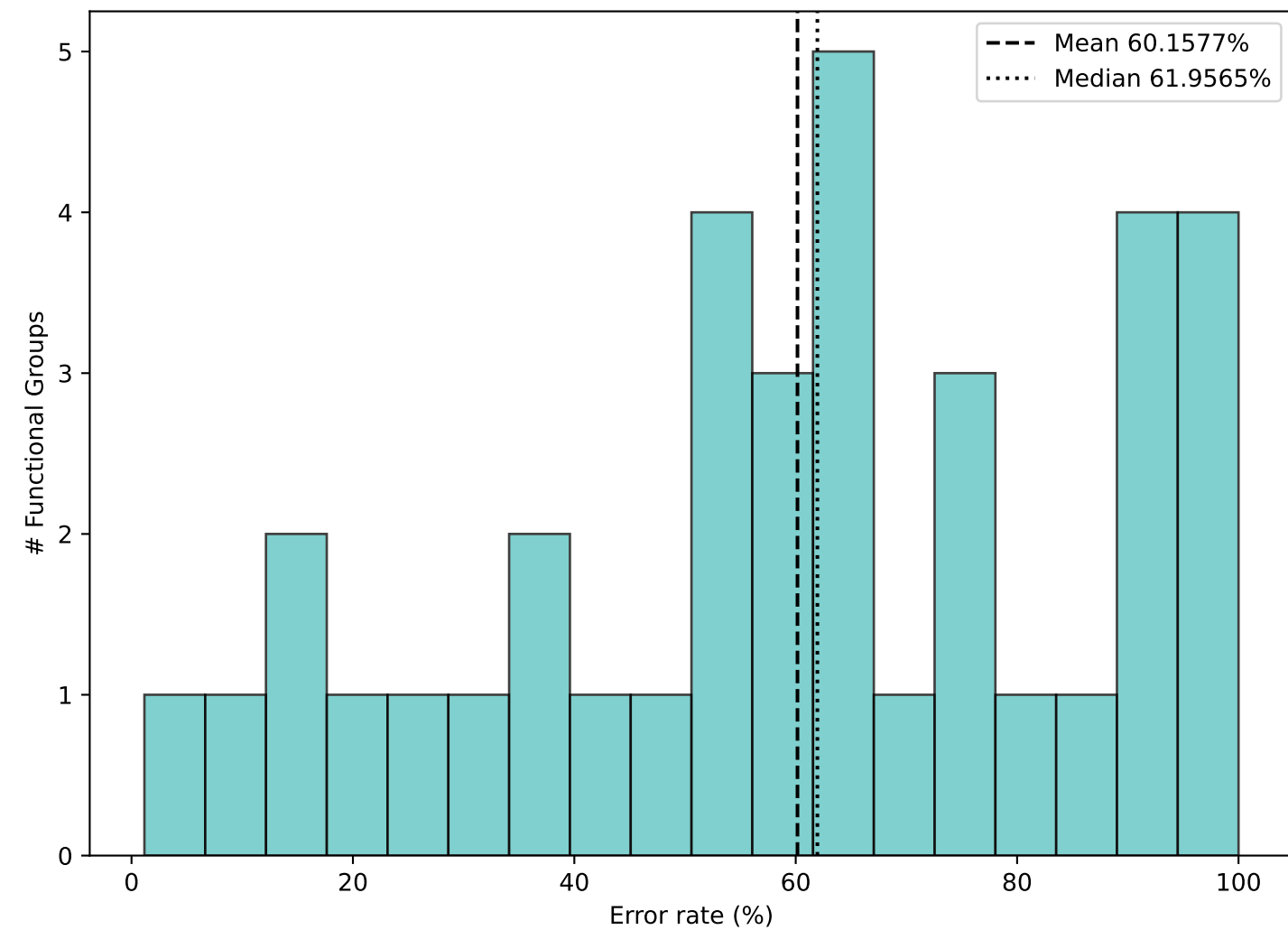


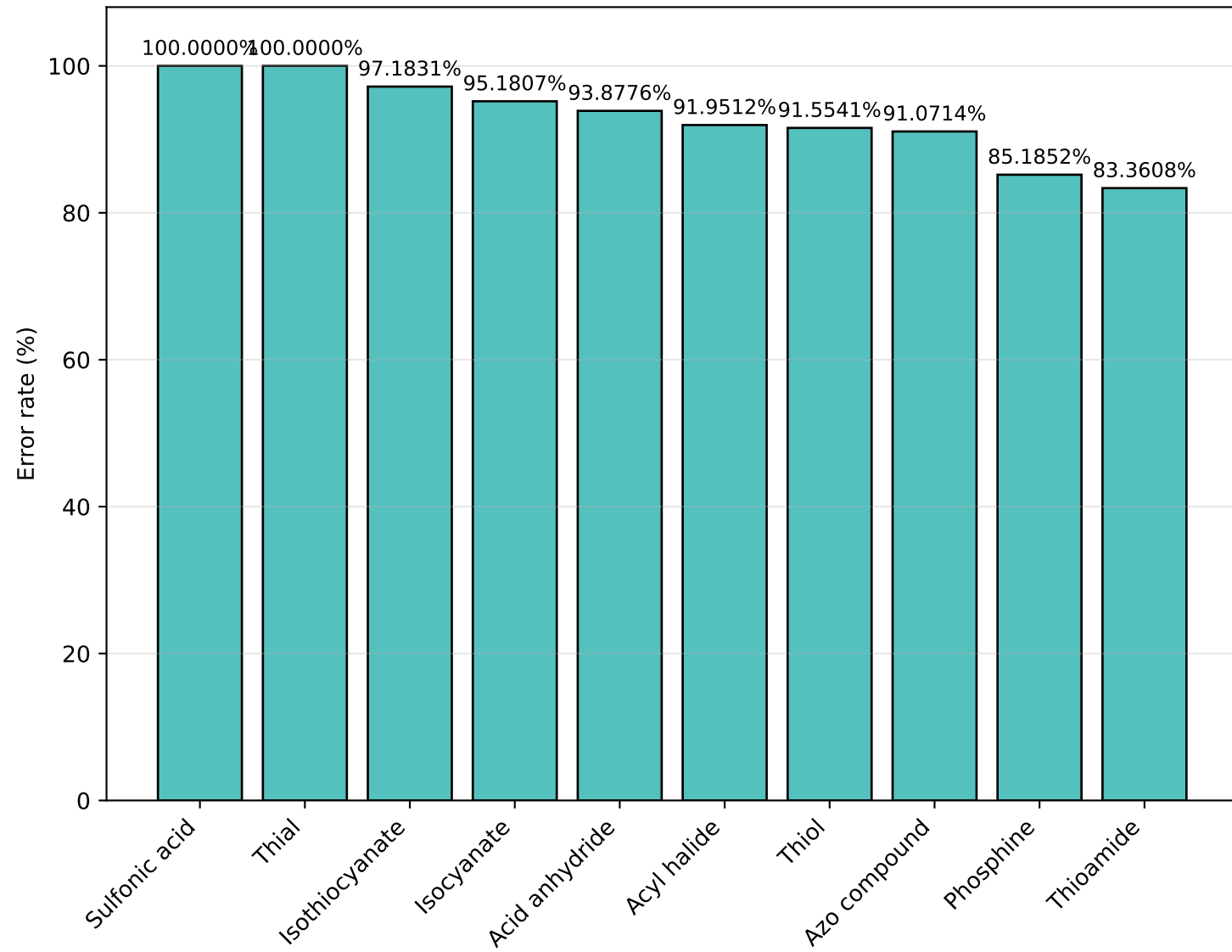
CNN H-NMR Validation Set - All 37 Functional Groups



Error-Rate Distribution



Top 10 Problem Groups



CNN H-NMR VALIDATION SET ERROR ANALYSIS

Global metrics:

F1 micro:	0.8234
F1 macro:	0.0436
Precision micro:	0.8889
Recall micro:	0.7669
Hamming accuracy:	0.9575

Error-rate statistics:

Mean:	60.1577%
Median:	61.9565%
Std:	27.5301%
Max:	100.0000% (Sulfonic acid)
Min:	1.1646% (Alkane)

Problem classes:

Severe (>50%):	26 groups
High (20-50%):	7 groups
Moderate (5-20%):	3 groups
Low (<=5%):	1 groups